# Notes: Brunetti, Harris, Mankad, and Michailidis (JFE, 2019)

**Interconnectedness in the Interbank Market**

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# 1 Big picture

- **Question.** How did bank interconnectedness evolve around the 2008 financial crisis, and do different network measures reveal different kinds of stress in the banking system?
- **Core distinction.** The paper compares two network structures:
  - **Correlation networks**: inferred from Granger-causality relations among publicly traded bank stock returns.
  - **Physical networks**: directly observed from interbank lending transactions on the European e-MID platform.
- **Main empirical fact.** Before the crisis, the two networks behave similarly. During the crisis, they move in opposite directions. Correlation-network interconnectedness rises, while physical interbank-trading interconnectedness falls.
- **Economic interpretation.** A rising correlation network means bank equity returns move together more strongly, consistent with common exposures and systemic-risk comovement. A falling physical network means banks trade less with each other, consistent with funding-market fragmentation, precautionary liquidity hoarding, and impaired interbank liquidity.
- **Setting and data.** The paper studies European banks from 2006 to 2012. The physical network uses 464,772 e-MID interbank transactions among 212 banks. The correlation network uses equity returns for the 54 publicly traded banks that can be matched to stock-market data.
- **Shock analysis.** The authors study how network statistics respond to macroeconomic shocks and ECB interventions/announcements. Physical networks react more strongly and quickly to ECB interventions, especially at short horizons, while correlation networks primarily capture crisis-level comovement in equity markets.
- **Forecasting result.** Both network types contain useful macro-financial information. Physical interbank networks are especially useful for forecasting several macro and sovereign-spread variables, which makes them valuable for regulators monitoring liquidity deterioration.
- **US–Europe transmission result.** Before Lehman, US bank correlation-network variables tend to Granger-cause European network variables. After Lehman, the direction more often reverses: European network variables tend to Granger-cause US variables.
- **Policy takeaway.** Correlation networks and physical networks are complements, not substitutes. Correlation networks are useful for detecting systemic comovement; physical networks are useful for monitoring the breakdown of funding liquidity and identifying early signs of interbank-market stress.

# 2 Data and measurement (key objects)

## 2.1 Sample period and crisis periods

The paper divides the sample into four periods:

- **Pre-crisis:** January 2, 2006–August 7, 2007.
- **Crisis 1, pre-Lehman:** August 8, 2007–September 12, 2008.

- **Crisis 2, post-Lehman:** September 16, 2008–April 1, 2009.
- **Crisis 3, post-recession/sovereign-stress period:** April 2, 2009–December 31, 2012.

**Interpretation.** The split isolates three conceptually different disruptions: the early funding crisis, the immediate Lehman shock, and the later European banking/sovereign-stress period.

### 2.2 Physical network: e-MID interbank trades

- **Source.** e-MID is an electronic regulated interbank market for euro-area unsecured deposits.
- **Observation.** Each transaction records the time, lender, borrower, interest rate, quantity, and the party initiating the trade.
- **Scale.** The sample contains 212 unique banks and 464,772 trades.
- **Frequency.** The paper constructs daily and monthly physical networks.
- **Edge definition.** If bank $B$ borrows from bank $A$ in the relevant time interval, draw a directed edge from $A$ to $B$.

**Interpretation.** The physical network directly measures realized interbank funding links. It is therefore a natural object for studying market liquidity, funding relationships, and hoarding behavior.

### 2.3 Correlation network: publicly traded bank returns

- **Source.** Equity returns for 54 publicly traded European banks.
- **Edge definition.** If the return of bank $A$ Granger-causes the return of bank $B$, draw a directed edge from $A$ to $B$.
- **Frequency.** The paper uses daily and monthly rolling windows.
- **Economic object.** The correlation network captures return comovement, common exposures, investor-perceived spillovers, and possibly contagion.

**Interpretation.** Correlation-network links are inferred from prices, not directly observed contracts. They may be very informative about systemic risk, but they cannot by themselves distinguish direct interbank transmission from common exposures.

### 2.4 Why the two networks differ

- The **physical network** is driven by banks' funding decisions.
- The **correlation network** is driven by investors' pricing of bank equity returns.
- The physical network can include private banks, but the correlation network can only include publicly traded banks.
- The physical network captures direct links; the correlation network captures direct and indirect statistical dependence.

**Key point.** A financial crisis can simultaneously increase stock-return comovement and decrease actual lending relationships. The two facts are not contradictory; they describe different margins of systemic stress.

# 3 Accounting framework and network construction (all equations + interpretation)

## 3.1 Bank balance sheet identity

Let bank $i$ hold assets across $k$ asset classes and have interbank liabilities, equity, and nonbank debt. The paper writes bank $i$'s balance-sheet identity as

$$\sum_k w_{i,k} y_{i,k} + \sum_j x_j \pi_{i,j} = e_i + x_i + d_i. \tag{1}$$

- $y_{i,k}$ is the market value of bank $i$'s asset class $k$.
- $w_{i,k}$ is bank $i$'s portfolio weight in asset $k$.
- $x_i$ is total liabilities of bank $i$ held by other banks.
- $x_{i,j}$ is the value of bank $i$'s liabilities held by bank $j$.
- $\pi_{i,j}$ is the share of bank $i$'s liabilities held by bank $j$.
- $e_i$ is the market value of bank $i$'s equity.
- $d_i$ is the value of bank $i$'s liabilities held by nonbanks.

A bank's assets consist of its portfolio holdings plus the interbank claims it holds on other banks. Its liabilities consist of equity, liabilities to other banks, and liabilities to nonbanks.

## 3.2 System representation of the interbank network

Stacking the balance sheets across banks gives

$$X = \Pi X + WY - E - D, \tag{2}$$

where $\Pi$ is the matrix of interbank liability shares. Equivalently,

$$(I - \Pi)X = WY - E - D. \tag{3}$$

**Interpretation.** The physical interbank market depends on banks' asset portfolios, bank equity values, and nonbank liabilities. If asset values, equity values, or funding needs move, the interbank network can adjust.

For the consolidated banking sector, interbank claims net out. The aggregate balance sheet is

$$E = WY - D. \tag{4}$$

**Interpretation.** Aggregation removes direct interbank claims. Equity-market correlations are therefore naturally linked to aggregate portfolio exposures and investor pricing, whereas physical-network links preserve direct lending relationships.

### 3.3 Correlation network construction

The stock return for bank $i$ is

$$r_{i,t} = \log\left(\frac{e_{i,t}}{e_{i,t-1}}\right).$$

After filtering returns using a GARCH(1,1) model, the authors run pairwise vector autoregressions. For a pair of bank returns collected in $U_t$,

$$\Phi(L)U_t = V_t, \tag{5}$$

where $V_t$ is normally distributed with covariance matrix $\Omega$. They test whether the off-diagonal lag terms are zero:

$$H_0 : \widehat{\Phi}(L) = 0. \tag{6}$$

**Read.** Rejecting the null means one bank's return helps predict another bank's return. That rejection creates a directed edge in the correlation network.

### 3.4 Physical network construction

For each day or month, define an adjacency matrix $A_t$ for e-MID trades:

$$A_{ij,t} = 1 \quad \text{if bank } i \text{ lends to bank } j \text{ in period } t,$$

and $A_{ij,t} = 0$ otherwise.

**Read.** Unlike the correlation network, this edge is directly observed. It is a transaction link, not an estimated statistical relation.

### 3.5 Network statistics

The paper uses five main measures of connectedness.

**Degree.**

$$\text{Degree} = \frac{1}{N(N-1)} \sum_{A=1}^{N} \sum_{B \neq A} A \to B. \tag{7}$$

Degree is the fraction of possible directed links that exist. In the physical network, lower degree means less interbank trading connectivity.

**Closeness.** Let $C_{AB}$ be the length of the shortest path from bank $A$ to bank $B$. Then

$$\text{Closeness} = \frac{1}{N(N-1)} \sum_{A=1}^{N} \sum_{B \neq A} C_{AB}. \tag{8}$$

Larger values indicate larger average distance between banks.

**Clustering coefficient.**

$$CC = \frac{3 \times \text{number of connected triples}}{\text{number of possible connected triples}}. \tag{9}$$

This measures how often a bank's neighbors are themselves connected.

**Eigenvector centrality.** This captures whether a bank is connected to other important banks. A bank is central if it links to banks that are themselves central.

**Largest strongly connected component.** LSCC is the proportion of nodes in the largest group where every bank can reach every other bank through directed paths.

**Interpretation.** These measures summarize different dimensions of network fragility: number of links, distance, clustering, centrality, and fragmentation.

# 4 Empirical specifications and identification (what makes the estimates credible)

## 4.1 Shock regressions

To study how network statistics respond to macroeconomic and ECB shocks, the paper estimates daily forecasting regressions. For network statistic $y_{t+k}$,

$$\begin{aligned} y_{t+k} = \alpha &+ \beta_1 U_t + \beta_2 S_t + \beta_3 DJST_t + \beta_4 \mathrm{EONIA}_t + \beta_5 (\mathrm{EURIBOR\text{-}OIS})_t \\ &+ \beta_6 \mathrm{LTRO}_t + \beta_7 \mathrm{MRO}_t + \beta_8 OT_t + \gamma y_{t-1} + \varepsilon_t. \end{aligned} \tag{10}$$

A related specification replaces the separate ECB operation dummies with a combined announcement variable:

$$\begin{aligned} y_{t+k} = \alpha &+ \beta_1 U_t + \beta_2 S_t + \beta_3 DJST_t + \beta_4 \mathrm{EONIA}_t + \beta_5 (\mathrm{EURIBOR\text{-}OIS})_t \\ &+ \beta_6 Announcements_t + \gamma y_{t-1} + e_t. \end{aligned} \tag{11}$$

- $U_t$ is an economic uncertainty index.
- $S_t$ is an economic surprise index.
- $DJST_t$ is the DJ Europe stock index.
- $\mathrm{EONIA}_t$ is the Euro Overnight Index Average.
- $(\mathrm{EURIBOR\text{-}OIS})_t$ proxies for banking-system stress.
- $\mathrm{LTRO}_t$, $\mathrm{MRO}_t$, and $OT_t$ capture ECB operations.
- $Announcements_t$ captures ECB conventional and unconventional announcements.

**Read.** The regression asks whether a given network statistic moves after macro-financial shocks or ECB policy actions. Estimating this separately for physical and correlation networks shows which information set each network is sensitive to.

## 4.2 F-tests and partial regressions

The paper tests whether macro shocks matter jointly:

$$H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0,$$

and whether ECB interventions matter jointly:

$$H_0 : \beta_6 = \beta_7 = \beta_8 = 0.$$

It also uses partial regressions to isolate ECB effects conditional on macroeconomic variables. Let $\widehat{y}_{t+k|1:5}$ be the fitted value from the macro-only regression. Then the ECB residual test is

$$y_{t+k} - \widehat{y}_{t+k|1:5} = \beta_0 + \beta_6 \mathrm{LTRO}_t + \beta_7 \mathrm{MRO}_t + \beta_8 OT_t + e_t. \tag{12}$$

**Read.** This asks whether ECB operations explain network movements over and above macro-financial shocks.

### 4.3 Forecasting macro-financial variables

The paper also asks whether network topology helps forecast macro-financial conditions. For macro variable $z_{i,t}$,

$$z_{i,t} = \gamma_0 + \gamma_1 Degree_{j,t-1} + \gamma_2 CC_{j,t-1} + \gamma_3 Closeness_{j,t-1} + \gamma_4 \mathrm{LSCC}_{j,t-1} + \gamma_5 z_{i,t-1} + u_{j,t}, \tag{13}$$

where $j$ denotes the network type: correlation or physical.

**Read.** If lagged network statistics forecast industrial production, retail sales, PMI, EURIBOR–OIS spreads, or sovereign spreads, then the network contains useful real-time monitoring information.

### 4.4 US–Europe transmission

Because the authors do not observe US interbank transactions, they construct US and European *correlation* networks and run Granger-causality tests between US and European network statistics.

**Read.** This tests whether systemic-risk comovement appears first in US bank equity networks or European bank equity networks, and whether the direction changes after Lehman.

## 5 Main results

### 5.1 Result 1: physical and correlation networks diverge during the crisis

- In the correlation network, degree, closeness, clustering, eigenvector centrality, and LSCC rise during the crisis.
- In the physical network, degree and LSCC decline strongly after the crisis begins.
- This means bank stock returns become more connected at the same time that actual interbank lending relationships become less connected.

**Interpretation.** Equity markets see banks as moving together because of common exposures and systemic distress. But the interbank market shows a breakdown in trading links, consistent with hoarding and distrust.

## 5.2 Result 2: different regions matter in different networks

- In correlation networks, banks from peripheral Europe contribute most to changes in interconnectedness.
- In physical networks, both core and peripheral European banks matter, but their importance changes over time.
- Core-country banks' importance in borrowing and lending reaches very low levels immediately after Lehman.

**Interpretation.** Return comovement concentrates around peripheral-bank stress, but actual interbank liquidity relationships depend on the evolving behavior of both core and peripheral banks.

## 5.3 Result 3: physical networks react more quickly to ECB interventions

- Physical-network statistics respond strongly to ECB operations and macroeconomic shocks during and after the crisis.
- Correlation-network statistics move less in response to ECB interventions, especially beyond the contemporaneous horizon.
- Conditional on macro shocks, ECB interventions still matter for the physical network at short horizons.

**Interpretation.** The physical network is closer to the funding-liquidity margin that ECB interventions directly target.

## 5.4 Result 4: networks forecast macro-financial conditions

- Both networks forecast hard information such as industrial production and retail sales.
- The physical network performs better for soft information such as PMI and for several sovereign spreads.
- Correlation networks and physical networks have similar explanatory power for the EURIBOR–OIS spread, but physical-network forecasts perform better for many variables.

**Interpretation.** Interbank trading networks are not merely descriptive. They can help regulators anticipate stress in the broader economy and banking system.

## 5.5 Result 5: post-Lehman transmission changes direction

- Before Lehman and during the early crisis, US bank correlation-network variables tend to Granger-cause European variables.
- After Lehman, European variables more often Granger-cause US variables.

**Interpretation.** Lehman changed not only the level of bank connectedness but also the geography of financial-stress transmission.

# 6 Tables and figures (original titles and original paper numbers)

All visuals below are extracted from the paper. The title shown above each image preserves the original figure/table number and original title/caption wording from the paper. Adjacent visuals are placed two side-by-side.

## 6.1 Raw returns and e-MID market variables

**Table 1. Summary statistics of the daily rates of stock returns ($\times 100$) for the different sub-periods.**

| Mean | Median | St. Dev. |
|---|---|---|
| Pre-crisis: 2-Jan-06 - 8-Aug-07 | | |
| 0.0603 | 0.0000 | 14.236 |
| Crisis 1: 9-Aug-07 - 12-Sep-08 | | |
| −0.1457* | −0.0886 | 15.449 |
| Crisis 2: 16-Sep-08 - 1-Apr-09 | | |
| −0.5037** | −0.1986 | 19.706 |
| Crisis 3: 2-Apr-09 – 31-Dec-12 | | |
| −0.0439 | 0.0000 | 19.849 |

*, **, and *** refer to significance levels of 10%, 5%, and 1% for testing the mean difference between each sub-period and the pre-crisis period that we use as benchmark. Standard errors are computed using boot-strapping.

**Table 2. Summary statistics of e-MID daily financial variables.**

| | Mean | Median | St. Dev. |
|---|---|---|---|
| **Pre-crisis: 2-Jan-06 - 8-Aug-07** | | | |
| Δ(Price) | −0.0232 | −0.0150 | 0.0871 |
| Effective spread | 1.3782 | 1.3888 | 0.0988 |
| Volume | 22,834 | 22,337 | 4902 |
| Trade imbalance | 0.0049 | 0.0046 | 0.0018 |
| Herfindahl index | 0.0159 | 0.0157 | 0.0014 |
| Signed volume | −13,154 | −12,715 | 5,6309 |
| **Crisis 1: 9-Aug-07 - 12-Sep-08** | | | |
| Δ(Price) | −0.1236*** | −0.0600 | 0.2224 |
| Effective spread | 1.3685 | 1.3804 | 0.1015 |
| Volume | 14,512*** | 14,132 | 3537 |
| Trade imbalance | 0.0067*** | 0.0064 | 0.0024 |
| Herfindahl index | 0.0173** | 0.0169 | 0.0022 |
| Signed volume | −8.7772*** | −8.5914 | 3.4672 |
| **Crisis 2: 16-Sep-08 - 1-Apr-09** | | | |
| Δ(Price) | −0.2832*** | −0.2500 | 0.2566 |
| Effective spread | 1.3629 | 1.3754 | 0.0939 |
| Volume | 7796*** | 7763 | 2568 |
| Trade imbalance | 0.0078*** | 0.0072 | 0.0027 |
| Herfindahl index | 0.0202*** | 0.0199 | 0.0026 |
| Signed volume | −4.3513*** | −4.0138 | 2.1801 |
| **Crisis 3: 2-Apr-09 - 31-Dec-12** | | | |
| Δ(Price) | −0.0070** | 0.0000 | 0.1945 |
| Effective spread | 1.3329 | 1.3465 | 0.1234 |
| Volume | 4002*** | 3997 | 1522 |
| Trade imbalance | 0.0199*** | −0.0190 | 0.0101 |
| Herfindahl index | 0.0521*** | 0.0484 | 0.0062 |
| Signed volume | −1.8971*** | −1.8283 | 1.4590 |

Trade imbalance is computed as the difference between number of buys and number of sells, normalized by volume. Signed volume is computed as the difference between aggressive buy volume and aggressive sell volume.
*, ** and, *** refer to significance levels of 10%, 5%, and 1% for testing the mean difference between each sub-period and the pre-crisis period. Standard errors are computed using bootstrapping.

**Read: Table 1.** Bank equity returns are positive before the crisis and become negative during the crisis periods. Volatility rises sharply, especially in the immediate post-Lehman period and remains elevated in the later European stress period.

**Read: Table 2.** e-MID trading volume collapses across the crisis periods. Average daily volume falls from more than 22,000 contracts pre-crisis to roughly 4,000 in Crisis 3. Trade imbalance and Herfindahl concentration rise, indicating a thinner and more concentrated interbank market.

## 6.2 Daily e-MID variables and monthly network summary

**Fig. 1. e-MID daily financial variables among 212 European banks from August 8, 2007, through December 31, 2012.**

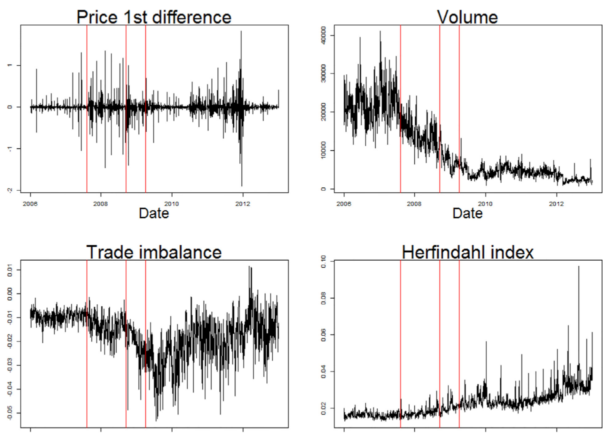



**Table 3. Summary statistics of monthly correlation and physical networks.**

| | Correlation network<br>Pre-crisis<br>2-Jan-06 - 8-Aug-07 | | | Physical network<br>Pre-crisis<br>2-Jan-06 - 8-Aug-07 | | |
|---|---|---|---|---|---|---|
| | Mean | Median | St. Dev. | Mean | Median | St. Dev. |
| Degree | 0.0419 | 0.0412 | 0.0048 | 0.0638 | 0.0632 | 0.0036 |
| Closeness | 0.0303 | 0.0303 | 0.0060 | 0.0053 | 0.0051 | 0.0004 |
| CC | 0.1459 | 0.1429 | 0.0247 | 0.3434 | 0.3405 | 0.0258 |
| EVCentrality | 0.0104 | 0.0095 | 0.0035 | 0.0170 | 0.0131 | 0.0090 |
| LSCC | 0.3411 | 0.3846 | 0.1666 | 0.4839 | 0.4858 | 0.0276 |
| | Crisis 1<br>9-Aug-07 - 12-Sep-08 | | | Crisis 1<br>9-Aug-07 - 12-Sep-08 | | |
| Degree | 0.0540*** | 0.0532 | 0.0180 | 0.0556*** | 0.0561 | 0.0050 |
| Closeness | 0.0386*** | 0.0388 | 0.0127 | 0.0051*** | 0.0051 | 0.0003 |
| CC | 0.1587 | 0.1527 | 0.0610 | 0.3512 | 0.3590 | 0.0263 |
| EVCentrality | 0.0147*** | 0.0133 | 0.0073 | 0.0113*** | 0.0091 | 0.0062 |
| LSCC | 0.4393*** | 0.5000 | 0.2345 | 0.4140*** | 0.4340 | 0.0378 |
| | Crisis 2<br>16-Sep-08 - 1-Apr-09 | | | Crisis 2<br>16-Sep-08 - 1-Apr-09 | | |
| Degree | 0.1061*** | 0.1081 | 0.0165 | 0.0415*** | 0.0393 | 0.0061 |
| Closeness | 0.0637*** | 0.0591 | 0.0087 | 0.0049*** | 0.0050 | 0.0004 |
| CC | 0.2761*** | 0.2780 | 0.0502 | 0.2881*** | 0.2795 | 0.0349 |
| EVCentrality | 0.0456*** | 0.0485 | 0.0119 | 0.0899*** | 0.0501 | 0.0781 |
| LSCC | 0.7088*** | 0.6923 | 0.0537 | 0.2358*** | 0.2170 | 0.0587 |
| | Crisis 3<br>2-Apr-09 - 31-Dec-12 | | | Crisis 3<br>2-Apr-09 - 31-Dec-12 | | |
| Degree | 0.1256*** | 0.1338 | 0.0261 | 0.0405*** | 0.0407 | 0.0049 |
| Closeness | 0.1029*** | 0.0894 | 0.0386 | 0.0051*** | 0.0051 | 0.0001 |
| CC | 0.3295*** | 0.3395 | 0.0693 | 0.3018*** | 0.3039 | 0.0330 |
| EVCentrality | 0.0583*** | 0.0641 | 0.0177 | 0.0238*** | 0.0143 | 0.0232 |
| LSCC | 0.7653*** | 0.7308 | 0.0824 | 0.1758*** | 0.1745 | 0.0457 |

Degree refers to the average degree in each network. Closeness measures the average distance, in terms of edges, between banks in the network. CC indicates the clustering coefficient. EVCentrality refers to the eigenvalue from eigenvector centrality, and LSCC refers to the proportion of nodes in the largest strongly connected component.
*, **, and *** refer to significance levels of 10%, 5%, and 1% for testing the mean difference between each sub-period and the pre-crisis period. Standard errors are computed using bootstrapping

**Read: Fig. 1.** Volume falls over time, trade imbalances become larger, and concentration rises. These patterns show that the physical interbank market becomes thinner and less competitive as the crisis develops.

**Read: Table 3.** Correlation-network connectedness rises strongly after the crisis begins: for example, degree rises from about 0.042 pre-crisis to about 0.126 in Crisis 3. Physical-network degree falls from about 0.064 pre-crisis to about 0.041 in Crisis 3, and physical LSCC falls from about 0.484 to about 0.176. This is the central empirical contrast of the paper.

## 6.3 Opposite network dynamics

**Fig. 2. Measures of interconnectedness (with statistics smoothed using local polynomial regression) for monthly physical networks for interbank lending among 212 European banks and correlation network statistics from 54 publicly traded European banks from August 8, 2007, through December 31, 2012.**

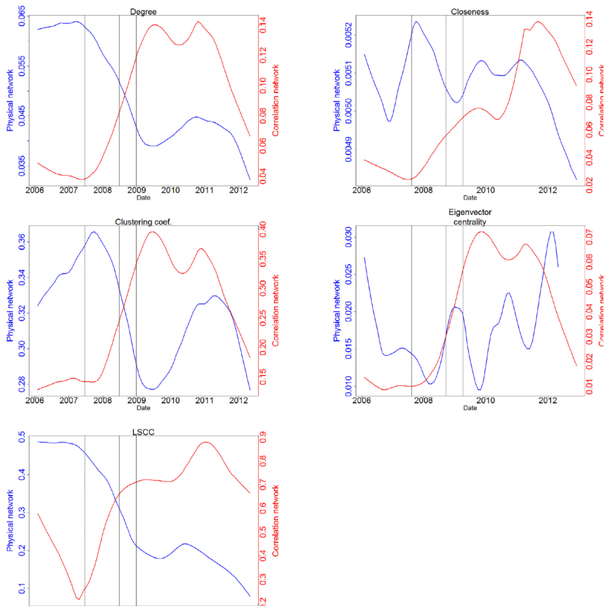
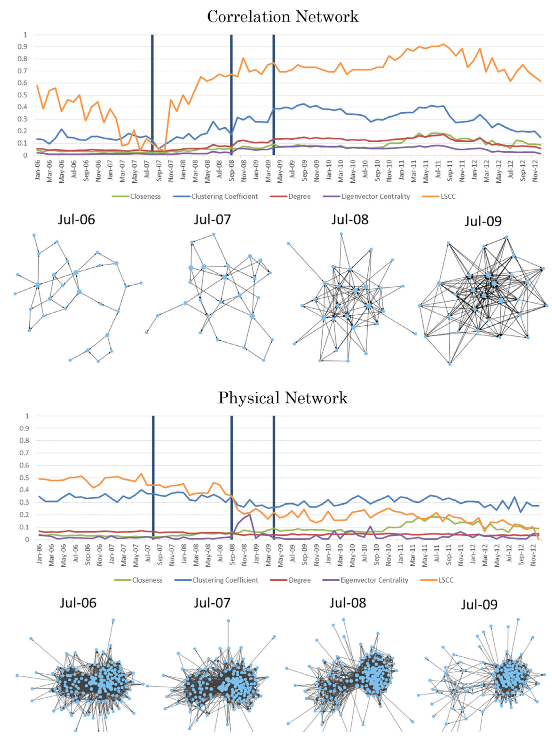



**Fig. 3. Time series of network statistics and corresponding graphs showing increasing connectivity in correlation networks and decreasing connectivity in physical networks.**




**Read: Fig. 2.** The two network types move in opposite directions. Correlation connectedness increases as stock returns move together, while physical connectedness decreases as banks stop trading with each other.

**Read: Fig. 3.** The graph visualizations make the same point visually: correlation-network graphs become denser, but physical-network graphs become more fragmented after the crisis begins.

## 6.4 Regional centrality and explanatory power of shocks

**Fig. 4. Directed bank (node) centrality measures (see Appendix) aggregated by geographic region.**

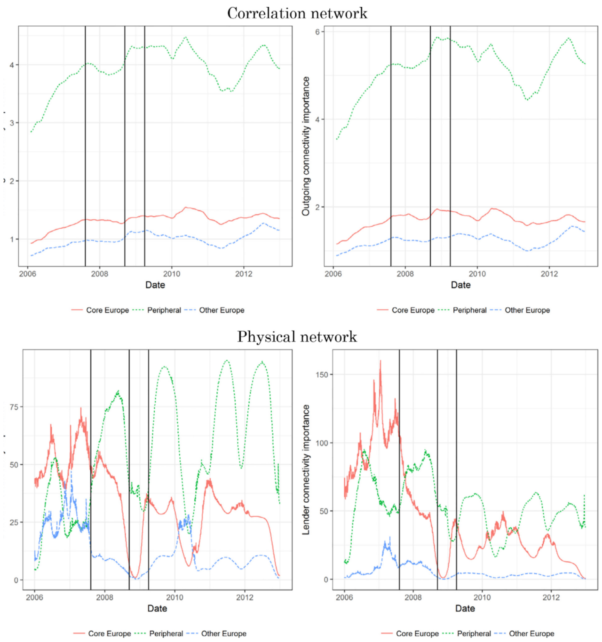
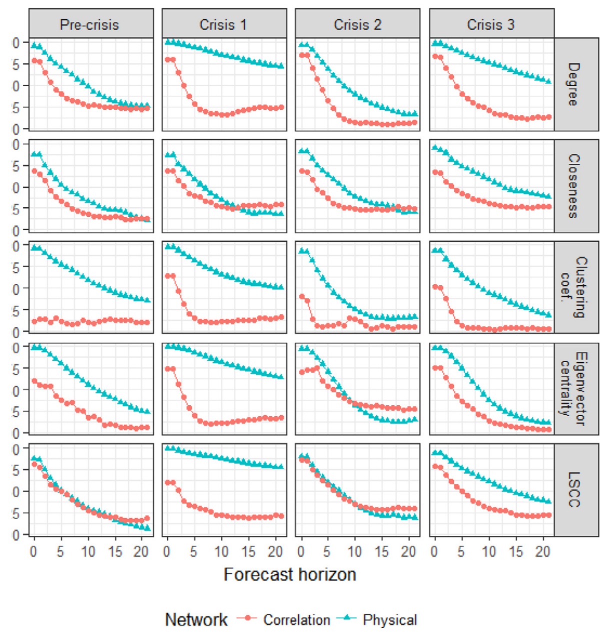



**Fig. 5. $R^2$ for the regressions in Eq. (12) showing that the physical network reacts more to ECB interventions and macroeconomic shocks during the crisis and following.**

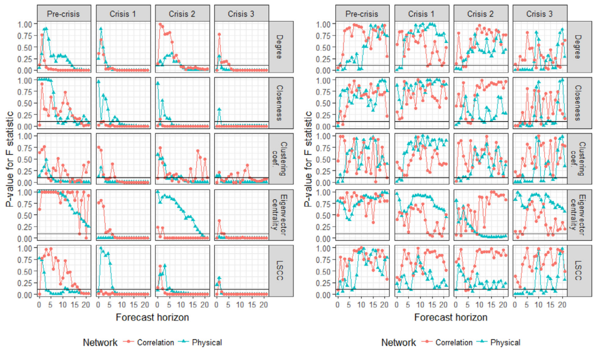



**Read: Fig. 4.** Peripheral-country banks are especially important in correlation networks. Physical-network centrality is more volatile, and core-country banks lose importance immediately after Lehman.

**Read: Fig. 5.** In the pre-crisis period, the two networks contain similar information. During and after the crisis, the physical network is more responsive to the shock variables, especially for several network statistics and forecast horizons.

## 6.5 Macro shocks, ECB shocks, and announcements

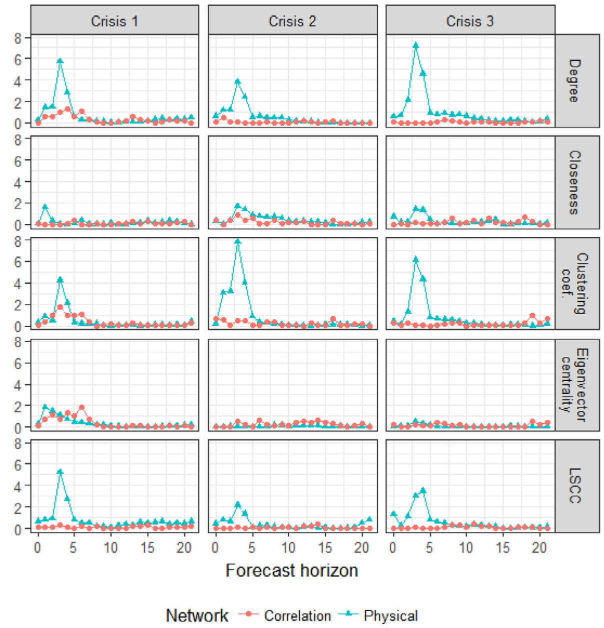
**Fig. 6. p-Values from F-tests for the regressions in Eq. (12).**




Macroeconomic shocks

ECB operations

**Fig. 7. Partial $R^2$ values of announcements conditional on macro shocks for the regressions in Eq. (13).**




**Read: Fig. 6.** Macro shocks matter for both networks, but the physical network is especially sensitive to ECB operations. Low p-values mean the relevant shock block significantly affects the network statistic.

**Read: Fig. 7.** ECB announcements add only modest incremental explanatory power once macro shocks are controlled for, and the additional effect tends to fade at longer horizons.

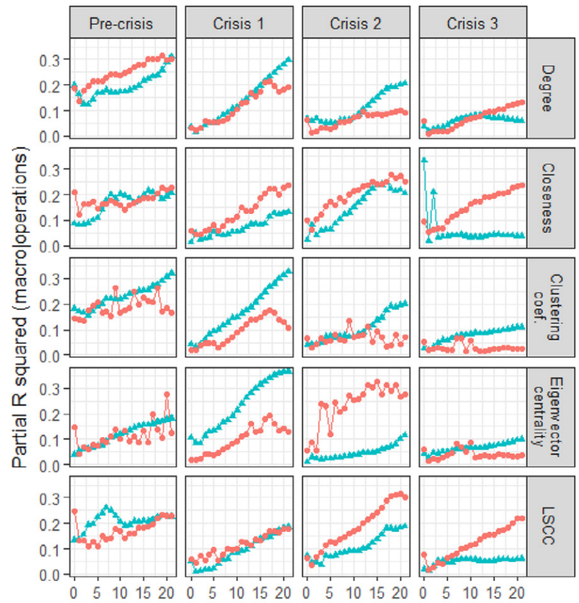
## 6.6 Forecasting macro-financial variables and conditional macro effects

**Table 4. Linear regression results of the network variables on the macro variables.**

| | $R^2$ | | | RMSE | | |
|---|---|---|---|---|---|---|
| | Correlation network | Physical network | Difference | Correlation network | Physical network | Difference |
| **Hard information** | | | | | | |
| $\Delta$(IP) | 0.3143 | 0.2763 | 0.0380 | 3.1867 | 2.9624 | 0.2243 |
| $\Delta$(RS) | 0.2146 | 0.1944 | 0.0202 | 1.9719 | 1.3710 | 0.6009* |
| **Soft information** | | | | | | |
| $\Delta$(PMI) | 0.3806 | 0.4730 | −0.0924** | 6.9828 | 4.3975 | 2.5853** |
| **Banking system health** | | | | | | |
| EURIBOR-OIS Spread | 0.6773 | 0.6739 | 0.0033 | 0.2255 | 0.7928 | −0.5673* |
| **Country-specific spreads** | | | | | | |
| ITSP | 0.0949 | 0.1795 | −0.0846** | 0.9620 | 0.8750 | 0.0870 |
| PTSP | 0.0190 | 0.1764 | −0.1574** | 1.1201 | 0.5976 | 0.5225* |
| GRSP | 0.1156 | 0.1922 | −0.0766** | 1.6393 | 1.4050 | 0.2343* |
| SPSP | 0.2777 | 0.1499 | 0.1279** | 1.2028 | 0.5798 | 0.6229* |

$R^2$ refers to the regression of the network variables (Degree, CC, Closeness, Eigenvector centrality, and LSCC) on the macro variables in the first column over the period January 2006 – December 2008. RSME refers to one-step-ahead forecasts from January 2009 until March 2010. Monthly observations. * and ** refer to significance levels of 10% and 5%. IP is industrial production, RS is retail sales, PMI is purchasing manager index, EURIBOR-OIS spread is the spread between the Euro Interbank Offered Rate and the Overnight Indexed Swap Rate, and ITSP, PTSP, GRSP, and SPSP represent spreads between the ten year Italian, Portuguese, Greek, and Spanish government bond yields and the ten year German government bond yields, respectively.

**Fig. 8. Partial $R^2$ values of macro shocks conditional on ECB interventions (operations) from the regressions in Eq. (12).**




**Read: Table 4.** Both networks forecast hard information, but the physical network has stronger forecasting performance for PMI and several sovereign spreads. This is important because it means interbank trading networks can serve as real-time monitoring tools.

**Read: Fig. 8.** Conditional on ECB interventions, macro shocks have persistent explanatory power for network statistics, especially at longer horizons.

### 6.7 Conditional ECB effects and US–Europe transmission

**Fig. 9. p-Values for the regressions in Eq. (15) corresponding to the null hypothesis that ECB interventions do not affect the network structure conditional on the macroeconomic variables.**

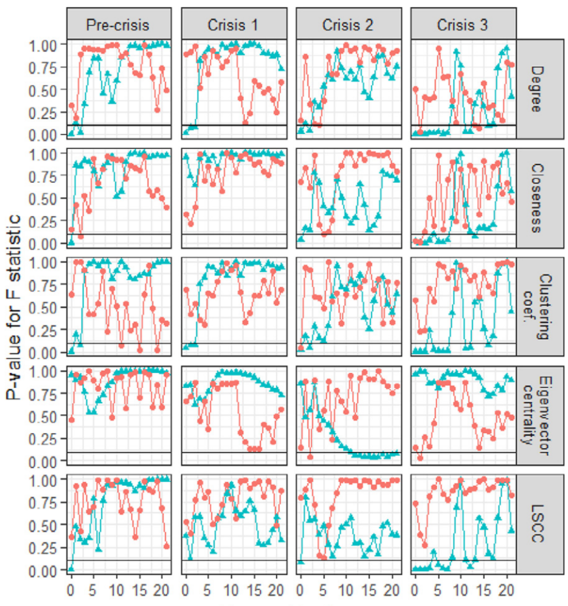



**Table 5. Granger causality results to measure the US–EU transmissions in network statistics.**

work statistics.

| | Degree | | |
|---|---|---|---|
| | US → EU | EU → US | Lags |
| Pre-crisis | 0.0006*** | 0.6340 | 3 |
| Crisis 1 | 0.0321*** | 0.1669 | 2 |
| Crisis 2 | 0.7613 | 0.0539*** | 1 |
| Crisis 3 | 0.5375 | 0.3265 | 4 |
| | Closeness | | |
| Pre-crisis | 0.5834 | 0.9045 | 2 |
| Crisis 1 | 0.5568 | 0.9577 | 2 |
| Crisis 2 | 0.5873 | 0.4738 | 2 |
| Crisis 3 | 0.0073*** | 0.0441*** | 4 |
| | Clustering coefficient | | |
| Pre-crisis | 0.0062*** | 0.4466 | 2 |
| Crisis 1 | 0.0783** | 0.7521 | 2 |
| Crisis 2 | 0.4872 | 0.0202*** | 2 |
| Crisis 3 | 0.7709 | 0.7507 | 4 |
| | Eigenvector centrality | | |
| Pre-crisis | 0.0618** | 0.1841 | 4 |
| Crisis 1 | 0.0105*** | 0.3144 | 3 |
| Crisis 2 | 0.7127 | 0.0455*** | 1 |
| Crisis 3 | 0.5778 | 0.6320 | 5 |
| | LSCC | | |
| Pre-crisis | 0.6127 | 0.6071 | 2 |
| Crisis 1 | 0.0949** | 0.7924 | 2 |
| Crisis 2 | 0.9612 | 0.0822** | 2 |
| Crisis 3 | 0.1480 | 0.0703** | 4 |

US → EU indicates that the US network Granger-causes the European network; EU → US indicates that the European network Granger-causes the US network; *, **, and *** refer rejection of the null of Granger-noncausality at significance levels of 10%, 5%, and 1%. The four subsamples are (1) pre-crisis period Jan 2, 2006 – Aug 7, 2007; (2) the first crisis period (pre-Lehman), Aug 8, 2007 – Sep 12, 2008; (3) the second crisis period (post-Lehman) Sep 16, 2008 – April 1, 2009; and (4)

**Read: Fig. 9.** ECB interventions explain physical-network movements even after conditioning on macro variables, especially at short horizons during crisis periods. This reinforces the idea that the physical network is the relevant liquidity-policy object.

**Read: Table 5.** Before Lehman, US bank network variables tend to lead European bank network variables. After Lehman, the direction more often reverses, with European variables leading US variables.

## 7 Short summary

- **Method.** Construct two bank networks: one from Granger-causality links in stock returns and one from observed e-MID interbank trades.
- **Core empirical result.** During the crisis, correlation networks become more connected, but physical interbank networks become less connected.
- **Interpretation.** Correlation networks measure systemic comovement and common exposures. Physical networks measure the health of direct funding relationships.
- **Shock result.** Physical networks respond more strongly and quickly to ECB interventions and macroeconomic shocks.

- **Forecasting result.** Network statistics forecast macro-financial variables; physical networks are particularly useful for predicting several stress-related variables.
- **Policy lesson.** Regulators should monitor both network types. Relying only on stock-return correlation networks can miss the breakdown of actual interbank liquidity.

# 8 Conclusion

**Core conclusion.** The paper shows that interconnectedness is not a single object. During the financial crisis, European banks became more connected in equity-return space but less connected in actual interbank lending space.

**Economic intuition.** In a crisis, investors price banks as sharing common risk, so stock-return correlation rises. At the same time, banks become reluctant to lend to each other, so the physical interbank network contracts. The same crisis therefore produces more statistical connectedness and less transactional connectedness.

**Takeaway.** Correlation networks are useful for systemic-risk monitoring, but physical networks are essential for understanding funding liquidity. A policy maker who observes only equity-return comovement may see systemic stress but miss the mechanism through which market liquidity is breaking down.